On Constrained Functional Extremization as the Unified Pricing Grammar of Gravity and Electromagnetism: A Meta-Unification Framework Based on Variational Isomorphism

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Abstract

This paper demonstrates that the Einstein field equations and Maxwell's equations do not need to be unified as the same kind of force—they are already two independent instantiations of the same unified pricing algorithm. This pricing algorithm is defined as "Li" (the price at which a transaction is finally reached under given constraints): under given constraints, a system automatically converges to the unique state that extremizes a certain functional among all possible states. The core argument proceeds in four steps: (i) gravity and electromagnetism cannot be unified at the level of "force"—their symmetry groups, field variable types, and action forms exhibit fundamental differences; (ii) however, at the meta-level, they share the same pricing grammar: both of their field equations are Euler-Lagrange equations derived from action extremization, sharing the identical extremum criterion $\delta S = 0$; (iii) more importantly, they exhibit precise structural isomorphism at the gauge field theory level—both gravity and electromagnetism are curvature equations of a connection, sharing the pricing grammar of "gauge connection $\rightarrow$ curvature $\rightarrow$ extremum criterion $\rightarrow$ field equation," with the former acting on the spacetime diffeomorphism group and the latter on the U(1) gauge group—this difference represents distinct implementations of the same algorithm under different constraints, not a difference in the algorithm itself; (iv) this grammar is not a coincidence internal to physics—classical mechanics, quantum mechanics, cosmology, chemical equilibrium and non-equilibrium thermodynamics, market clearing and financial instability, channel capacity and complexity threshold, and entropy increase—the core pricing rules of these disciplines were independently derived throughout history, by researchers unaware of each other's work, using different mathematical tools, yet ultimately converging on the same constrained extremization skeleton in mathematical structure. Independently derived isomorphism is more convincing than intentionally inherited isomorphism. This paper precisely situates Li's unification framework within the history of ideas through systematic comparison with previous unification attempts, and honestly marks the boundary of the theory: in the arena of science, the theory remains at "all things conform to the

description of Li"; under the stars of philosophy, the theory retains the ontological conviction that "all things are themselves the unfolding of Li"—but conviction is not a proven theorem. The highest is not conquest; the highest is honesty.
Keywords: unified field theory; variational principle; constrained extremization; functional analysis; gauge field theory; interdisciplinary unification; incompleteness theorem; complexity threshold

1 Introduction: The Historical Dilemma of Unified Field Theory and the Necessity of Meta-Unification

From Einstein to string theory, one of the core goals of theoretical physics has been to unify gravity and electromagnetism within a single mathematical framework. Einstein spent the latter half of his life attempting to find a unified geometric structure [1], but never succeeded. All previous unification attempts sought unification at the level of "force"—attempting to reduce both to a single, more fundamental entity.

This direction faces fundamental difficulties. Gravity and electromagnetism exhibit irreconcilable differences in physical properties, symmetry groups, and action forms: (i) the source of gravity is mass, producing only attraction; the source of electromagnetism is charge, which can produce attraction or repulsion; (ii) gravity is described by a second-rank tensor field (spin-2), while electromagnetism is described by a vector field (spin-1); (iii) gravity is a manifestation of spacetime geometry, with the diffeomorphism group as its symmetry group; electromagnetism is a gauge field, with the U(1) gauge group as its symmetry group. The two cannot be compressed into a single "force" formula.

This paper proposes that unification need not be accomplished at the level of "force." Gravity and electromagnetism share a deeper pricing grammar—constrained functional extremization. This is a meta-level unification: not compressing two forces into a single formula, but recognizing that they are already two different instantiations of the same formula. The structure of this paper's argument is as follows: Sections 3 through 5 complete the meta-unification proof for gravity and electromagnetism, Sections 6 through 7 demonstrate the interdisciplinary universality of this unification, Section 8 provides precise positioning within the history of ideas, and Section 9 marks the boundary and the honest distinction of ontological conviction.

2 Formal Definition of the Unified Pricing Algorithm

2.1 The Triadic Structure of the Algorithm

Definition 2.1 (Unified Pricing Algorithm)**.** *Let the pricing algorithm of a physical system under given constraints be the triple* $(\mathcal{S}, \mathcal{C}, \delta)$*, where:*

- $\mathcal{S}$ *is the state space, the set of all mathematically possible field configurations;*
- $\mathcal{C}$ *is the constraint conditions, restricting* $\mathcal{S}$ *to the physically possible subspace* $\mathcal{S}_{\mathcal{C}} \subset \mathcal{S}$*;*
- δ *is the extremum criterion, selecting the uniquely realized field configuration from* $\mathcal{S}_{\mathcal{C}}$*.*

In physics, the extremum criterion manifests as the extremization of an action functional. Let the action functional be

$$S[\phi] = \int_{\mathcal{M}} \mathcal{L}(\phi, \partial\phi, x)\, d^n x \tag{1}$$

where ϕ is the field variable, $\mathcal{L}$ is the Lagrangian density, and $\mathcal{M}$ is the base manifold. The extremum criterion is

$$\delta S[\phi] = 0. \tag{2}$$

This criterion provides a rigorous mathematical mechanism for selecting the unique realization from among possible states. The constraint conditions $\mathcal{C}$ can be further classified into three types: (i) geometric constraints—the topological and metric structure of the base manifold; (ii) internal constraints—the symmetry requirements of the field (gauge invariance, Lorentz invariance); (iii) boundary constraints—the behavior of the field at infinity or at initial moments. Constraint conditions determine the boundary of the possible state space, within which the extremum criterion locates the unique extremum point.

It should be noted that the core argument of this paper is based on the extremum criterion $\delta S = 0$, and does not depend on whether the extremum is a minimum, maximum, or saddle point. The constraint conditions select from all possible configurations the configuration that makes the first variation of the functional vanish—this configuration is the unique output of the pricing algorithm. For discussion of sufficient conditions for extrema (Hilbert's condition), see Appendix B.

2.2 Definition and Interpretation of Li

The general form of this pricing algorithm can be stated as follows: any system, under the interactive tendencies and capacity constraints of its internal elements, through continuous interaction, will gradually converge to a unified equilibrium state; during the formation of this equilibrium state, the system necessarily and synchronously accumulates structural tensions that cannot be fully accommodated by the current equilibrium; when such tensions accumulate beyond the system's bearable threshold, the system will ultimately be triggered to reshape its equilibrium structure, or the equilibrium structure will disintegrate. For ease of exposition, this paper names this algorithm "Li", defined as the price at which a transaction is finally reached under given constraints.

The everyday language formulation of Li is: in a marketplace, after bargaining among all parties, the price at which a transaction is finally reached—that is Li. This formulation is fully equivalent to the formal definition: "all parties" corresponds to the multiple possible configurations in the state space; "bargaining" corresponds to the interaction under constraint conditions; "the price at which a transaction is finally reached" corresponds to the unique realization selected by the extremum criterion. "Finally" rather than "uniquely"—"finally" is the unfolding of "uniquely" in time: it is not a static mathematical statement, but the equilibrium point to which the system automatically converges after the constraint conditions drive it through an unskippable bargaining process.

3 Proof of Gravity's Unified Pricing

The Einstein-Hilbert action (without cosmological constant) is

$$S_{\mathrm{EH}}[g_{\mu\nu}] = \frac{1}{2\kappa}\int_{\mathcal{M}} R\sqrt{-g}\, d^4x \tag{3}$$

where $g_{\mu\nu}$ is the metric field (gravitational potential), R is the Ricci scalar (spacetime curvature), g is the determinant of the metric, $\kappa = 8\pi G/c^4$ is the Einstein gravitational constant, and $\mathcal{M}$ is the four-dimensional spacetime manifold.

Varying with respect to the metric $g_{\mu\nu}$ and requiring $\delta S_{\mathrm{EH}} = 0$ yields the vacuum Einstein field equations

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 0. \tag{4}$$

Adding the matter Lagrangian density $\mathcal{L}_{\mathrm{matter}}$ yields the full Einstein field equations

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \kappa T_{\mu\nu} \tag{5}$$

where $T_{\mu\nu}$ is the matter energy-momentum tensor. The field equations are the Euler-Lagrange form of the extremum criterion.

The physical configuration of the gravitational field is not arbitrarily given. Among all mathematically possible metric fields, only the one that extremizes the Einstein-Hilbert action is realized. Constraint conditions include: the four-dimensional spacetime manifold is a fixed base space (geometric constraint); the metric tensor must be symmetric and possess a Lorentzian signature (internal constraint); the metric and its first derivatives are fixed on the boundary (boundary constraint). The extremum criterion is $\delta S_{\mathrm{EH}} = 0$. Gravity is the process by which spacetime geometry prices its own curvature under constraint conditions.

4 Proof of Electromagnetism's Unified Pricing

The Maxwell action (with source) is

$$S_{\mathrm{EM}}[A_\mu] = \int_{\mathcal{M}} \left(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu\right) d^4x \tag{6}$$

where A_μ is the electromagnetic four-potential, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field tensor, and J^μ is the four-current density.

Varying with respect to A_μ and requiring $\delta S_{\mathrm{EM}} = 0$ yields the Maxwell equations with source

$$\partial_\mu F^{\mu\nu} = J^\nu. \tag{7}$$

In the absence of sources, this reduces to the homogeneous Maxwell equations $\partial_\mu F^{\mu\nu} = 0$.

The physical configuration of the electromagnetic field is not arbitrarily given. Among all mathematically possible four-potential fields, only the one that extremizes the Maxwell action is realized. Constraint conditions include: the base manifold is four-dimensional spacetime (geometric constraint); gauge invariance $A_\mu \to A_\mu + \partial_\mu\Lambda$ (internal constraint); boundary conditions are fixed (boundary constraint). The extremum criterion is $\delta S_{\mathrm{EM}} = 0$. Electromagnetism is the process by which the electromagnetic field prices its own field configuration under constraint conditions.

5 Isomorphism Proof: Not the Same Kind of Force, but the Same Pricing Algorithm

5.1 Strict Distinction Between Two Kinds of "Unification"

Definition 5.1 (Same Kind of Force)**.** *Two theories describe the same kind of force if and only if they share the same physical mechanism and are mutually reducible under appropriate limits. Gravity and electromagnetism do not satisfy this definition: the source of gravity is mass (pure attraction), while the source of electromagnetism is charge (attraction or repulsion); gravity is described by a second-rank tensor field (spin-2), while electromagnetism is described by a vector field (spin-1); gravity is a manifestation of spacetime geometry, while electromagnetism is a gauge field.*

Definition 5.2 (Same Pricing Algorithm)**.** *Two theories share the same pricing algorithm if and only if (i) both are instances of functional extremization problems; (ii) both share the extremum criterion $\delta S = 0$; (iii) the field equations of both satisfy the grammar of "constraint conditions determine the space of possible states, extremum criterion selects the unique realization from among them." Gravity and electromagnetism strictly satisfy this definition.*

5.2 Isomorphism at the Variational Level

Let the state space of the gravitational system be $\mathcal{G}$ (the set of all possible metric fields), and the state space of the electromagnetic system be $\mathcal{A}$ (the set of all possible four-potential fields). The constraint conditions $\mathcal{C}_{\text{EH}}$ and $\mathcal{C}_{\text{EM}}$ respectively restrict these two spaces to physically possible subsets. The extremum criterion $\delta S = 0$ selects from these subsets the uniquely realized physical field configuration.

The two systems are not completely identical: $\mathcal{G}$ is a second-rank tensor field space, while $\mathcal{A}$ is a vector field space; the symmetry group of gravitational action is the spacetime diffeomorphism group, while the symmetry group of electromagnetic action is the U(1) gauge group. However, both share the same base manifold (four-dimensional spacetime manifold), the same extremum criterion ($\delta S = 0$), and their Euler-Lagrange equations are instances of the same skeleton under different field variables.

5.3 Isomorphism at the Gauge Field Theory Level

Gravity and electromagnetism are not merely two independent variational problems; they exhibit precise structural isomorphism at the level of gauge field theory. For any gauge theory, so long as its action is the curvature squared (or curvature scalar) of a connection, its field equations are necessarily the pricing equation of curvature with source terms. What gravity and electromagnetism share is not the same gauge group, but the same pricing grammar of "gauge connection $\to$ curvature $\to$ extremum criterion $\to$ field equation."

In gravitational theory, the Levi-Civita connection $\Gamma^{\lambda}_{\mu\nu}$ is the gauge connection of spacetime, and the Ricci tensor $R_{\mu\nu}$ is the curvature of this connection—it measures the deflection of a vector after parallel transport around a closed path. The Einstein field equations are the pricing equation of Ricci curvature with matter distribution. In electromagnetic theory, the four-potential A_{μ} is the gauge connection of electromagnetism, and the field tensor $F_{\mu\nu}$ is the curvature of this connection—it measures the deflection

Table 1: Structural Isomorphism between Gravity and Electromagnetism

Structural Element	**Gravity**	**Electromagnetism**
Gauge connection	Levi-Civita connection $\Gamma^{\lambda}_{\mu\nu}$	Four-potential A_μ
Curvature	Ricci tensor $R_{\mu\nu}$	Field tensor $F_{\mu\nu}$
Curvature equation	$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \kappa T_{\mu\nu}$	$\partial_\mu F^{\mu\nu} = J^\nu$
Action	$S_{\text{EH}} = \frac{1}{2\kappa}\int R\sqrt{-g}\, d^4x$	$S_{\text{EM}} = \int(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu})\, d^4x$
Extremum criterion	$\delta S_{\text{EH}} = 0$	$\delta S_{\text{EM}} = 0$

of phase after transport around a closed path. The Maxwell equations are the pricing equation of this curvature with current density.

The deeper theoretical reason for this isomorphism is that *both connections are the unique torsion-free, metric-compatible connections on their respective fiber bundles*—the Levi-Civita connection on the tangent bundle for gravity, and the Ehresmann connection on a principal U(1)-bundle for electromagnetism. In each case, the connection is fully determined by the geometric structure of the underlying bundle and the requirement of compatibility with a given structure group. The curvature of the connection then automatically satisfies Bianchi identities—$\nabla_{[\lambda}R_{\mu\nu]\rho\sigma} = 0$ for gravity and $\partial_{[\lambda}F_{\mu\nu]} = 0$ for electromagnetism—which are direct consequences of the connection being derived from a potential. These identities are not independent physical laws superimposed on the theory; they are mathematical necessities arising from the very definition of curvature as the commutator of covariant derivatives. Furthermore, in both theories, the action functional is the unique gauge-invariant scalar of lowest mass dimension constructible from the curvature: for gravity, the Ricci scalar R is the unique such scalar from the Riemann tensor; for electromagnetism, $F_{\mu\nu}F^{\mu\nu}$ is the unique such scalar from the field tensor. Thus, both theories are not merely analogous—they are mathematically inevitable: given a connection on a fiber bundle with a structure group, the curvature, the Bianchi identities, and the unique lowest-order gauge-invariant action all follow deductively. That gravity and electromagnetism converge on the same pricing grammar of "connection → curvature → extremum criterion → field equation" is therefore not a coincidental formal parallel, but a necessary consequence of the underlying differential geometry of gauge theories on fiber bundles.

5.3.1 Gauge Invariance as the Mechanism of Pricing Grammar Determination

The identification of gauge invariance as the underlying mechanism determining the pricing grammar reveals a crucial insight: *the pricing grammar of a physical theory is not chosen by nature from a menu of possibilities—it is uniquely dictated by the requirement of gauge invariance under a given symmetry group.* For gravity, diffeomorphism invariance (the gauge invariance of general relativity) uniquely selects the Einstein-Hilbert action as the lowest-order term in an expansion of curvature invariants. For electromagnetism, U(1) gauge invariance uniquely selects the Maxwell action as the lowest-order gauge-invariant term. In both cases, the action—and therefore the entire pricing grammar—is the unique solution to the constraint of gauge invariance. This is precisely the logic of Li: *constraint conditions (here, gauge invariance) determine the space of possible actions; the extremum criterion ($\delta S = 0$) selects the unique physical realization from that space.* The unified pricing algorithm does not merely describe gauge theories—it is *instantiated*

by gauge theories at the most fundamental level of their mathematical structure.

It should be noted that the above isomorphism argument is rigorous at the classical field theory level. Electromagnetism has been successfully quantized as quantum electrodynamics (QED), where its quantum-level pricing grammar is expressed by the Feynman path integral—a functional integral over all possible field configurations, with the extremum path emerging through stationary phase approximation in the $\hbar \to 0$ limit. The quantization of gravity (quantum gravity) remains an unsolved problem in physics—it is not yet certain whether quantum gravity obeys a pricing grammar fully isomorphic to that of quantum electromagnetism. This unresolved issue does not affect the core argument of this paper at the classical field theory level, but it is a critical checkpoint for whether Li's pricing grammar can extend from the classical domain to the quantum gravity domain.

5.4 Compatibility of Symmetry Group Differences

The symmetry group of gravity is the diffeomorphism group (external symmetry of spacetime), while the symmetry group of electromagnetism is the U(1) gauge group (internal symmetry of the field). The two are incompatible at the group-theoretic level and cannot be mutually reduced. This difference represents distinct implementations of the pricing algorithm under different constraint conditions, not a difference in the algorithm itself.

5.5 Anticipatory Responses to Potential Criticisms

A rigorous critic might point out that almost all classical field theories obey variational principles, and therefore "sharing the variational principle" cannot constitute a meaningful unification—this would make any two field theories instances of "unified pricing." This paper offers three responses.

First, the unification proposed in this paper is not based on the superficial commonality of "both obeying variational principles," but on the deep structural isomorphism of "both being curvature equations of a connection"—not all field theories share this structure. For example, the action of scalar field theory is the Klein-Gordon action $S_{\text{KG}} = \int (-\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2)d^4x$, whose field equation $\partial_\mu\partial^\mu\phi - m^2\phi = 0$ is a wave equation rather than a curvature equation of a connection. Scalar field theory also obeys the variational principle, but it is not a gauge field theory. Therefore, "sharing the variational principle" is a property common to all classical field theories, but "sharing the pricing grammar of gauge connection $\to$ curvature $\to$ extremum criterion $\to$ field equation" is an isomorphism specific to gravity and electromagnetism.

Second, and more importantly, this paper further demonstrates in Section 6 that this skeleton independently holds across multiple disciplines beyond physics—if this skeleton were merely a formal coincidence internal to physics, it could not have been independently derived in chemistry, economics, information science, and energy science as well.

Third, regarding this paper's focus on gravity and electromagnetism rather than all four fundamental forces: the weak and strong forces, as gauge field theories, also obey variational principles, and they are consistent with gravity and electromagnetism in the basic structure of the pricing grammar. The weak and strong forces acquire mass through the Higgs mechanism, while the photon and graviton (theoretically) are massless—this difference represents distinct implementations of the pricing algorithm under different gauge groups and symmetry-breaking constraints. This paper focuses on gravity and

electromagnetism because they are long-range forces, and the historical dilemma of Einstein's unified field theory is most concentratedly embodied in the unification of these two forces. The complete formal analysis of the weak and strong forces within Li's framework is left for future research.

The above responses constitute the closure of the core argument. The following Sections 6 through 7 demonstrate that this unification is not a coincidence internal to physics, but a universal underlying grammar across disciplines.

6 Interdisciplinary Corroboration: The Universality of the Unified Pricing Grammar

The above has demonstrated that gravity and electromagnetism share the same pricing grammar at the meta-level. This section further demonstrates that this grammar is not a coincidence internal to physics. The core pricing rules of the following eight disciplines were independently derived throughout history, by researchers unaware of each other's work, using different mathematical tools, yet ultimately converged on the same constrained extremization skeleton in mathematical structure.

6.1 The Principle of Least Action in Classical Mechanics

Action $S = \int (T - V)dt$, extremum criterion $\delta S = 0$. Newton (1687) derived the laws of motion from the instantaneous equilibrium of forces [2], while Euler (1744) independently derived the brachistochrone from the variational principle [3]—the two were unaware of each other's work, yet arrived at the same destination by different paths.

6.2 The Path Integral in Quantum Mechanics

Feynman path integral [4] $K(b, a) = \int \mathcal{D}[x(t)]e^{iS/\hbar}$. When $S \gg \hbar$, paths near $\delta S = 0$ are enhanced by stationary phase approximation to become the classical limit.[1] Heisenberg's uncertainty principle [5] $\Delta x \cdot \Delta p \geq \hbar/2$ stipulates that conjugate constraint variables cannot be simultaneously measured with precision. Feynman's path integral and Heisenberg's matrix mechanics were independently derived and only later proven equivalent.

6.3 The Friedmann Equation in Cosmology

$H^2 = (8\pi G/3)\rho - kc^2/a^2 + \Lambda c^2/3$ [6]. Friedmann derived this equation from Einstein's field equations in 1922; Hubble independently discovered cosmic expansion from observation in 1929 [7]—theoretical derivation and observational verification were unaware of each other.

[1] The mathematical rigor of the path integral—a functional integral over infinite-dimensional space—remains an active research area in mathematical physics, but this does not affect the predictive power of the path integral in physics or the argument of this paper.

6.4 Gibbs Free Energy Minimization and Activation Energy in Chemistry

Gibbs free energy criterion [8] $\Delta G \leq 0$, with $\Delta G = 0$ at equilibrium. The activation energy constraint [9] $k = A\exp(-E_a/RT)$ stipulates that only the path with the lowest activation energy is realized. These two were independently derived and together constitute the complete pricing grammar of chemistry.

6.5 Dissipative Structures in Non-Equilibrium Thermodynamics

Entropy production rate $\sigma = \sum J_i X_i \geq 0$. Onsager reciprocal relations [10] $L_{ij} = L_{ji}$ (1931) stipulate linear pricing in the near-equilibrium regime; dissipative structure theory [11] (1971) stipulates nonlinear pricing in the far-from-equilibrium regime—research in these two phases was unaware of each other.

6.6 Market Clearing and Financial Instability in Economics

Walrasian equilibrium [12] $D(p^*) = S(p^*)$, with the Arrow-Debreu proof [13] of its existence and Pareto efficiency. Minsky [14] further demonstrated that the same pricing algorithm sequentially outputs hedge, speculative, and Ponzi prices under the dynamic changes of financing constraints, ultimately outputting crash prices—the Minsky moment manifests mathematically as a rapid transition of the system from one stable attractor to another, a classical phenomenon of bifurcation theory, structurally isomorphic to non-equilibrium phase transitions in dissipative structure theory. Smith's "invisible hand" and Minsky's "clenching hand" are the same hand in different forms under two different market constraint conditions. These two were independently derived, using different mathematical tools—fixed-point theorems and macro-dynamics.

6.7 Channel Capacity and Complexity Threshold in Information Science

Shannon channel capacity [15] $C = B\log_2(1 + S/N)$ (1948).[2] Von Neumann complexity threshold [16] $N_{\text{crit}} = N_0 e^{\alpha k}$ (1956). The two were unaware of each other's work, yet their mathematical structures were later discovered to be fully isomorphic: both are exponential growth, irreversible, and involve critical thresholds.

6.8 The Second Law of Thermodynamics in Energy Science

Carnot efficiency [17] $\eta = 1 - T_2/T_1$ (1824). Clausius entropy increase principle [18] $dS \geq \delta Q/T$ (1865). The constant discrepancy repeatedly measured by Faraday [19] in his electromagnetic induction experiments—not error, not insufficient precision of equipment, but entropy. Carnot independently derived the efficiency limit in 1824, at a time when the concept of entropy had not yet been proposed.

[2]Shannon's channel capacity formula assumes additive white Gaussian noise. For non-Gaussian noise, the general form of channel capacity is $C = \sup_{p(x)} I(X;Y)$—this remains a constrained functional extremization problem. The structure of the pricing grammar is unchanged; only the specific forms of the constraint conditions (noise distribution) and the extremum criterion (mutual information maximization) differ.

7 Comprehensive Proof of Isomorphism: Li as a Cross-Disciplinary Unified Grammar

The core formulas of the above eight disciplines are, in mathematical structure, the same thing—different instances of constrained functional extremization problems: $\delta S = 0$, $\Delta G = 0$, $D(p) = S(p)$, $C = \sup I(X;Y)$, $dS \geq \delta Q/T$. Their discoverers were unaware of each other, using different mathematical tools—calculus of variations, differential equations, fixed-point theorems, convex optimization, functional analysis. That they ultimately coincide in mathematical structure is not because later generations deliberately unified them, but because each independently converged to the same skeleton under its own constraint conditions. Independently derived isomorphism is more convincing than intentionally inherited isomorphism—it implies that this skeleton is the unique solution to which each discipline automatically converges under constraint conditions, not a deliberate construction by any individual.

This paper does not "invent" this unification—this paper "recognizes" that they are already the same thing. Constraint conditions determine the space of possible states; the extremum criterion selects the unique realization from among possible states. This grammar runs through all levels of existence, from quanta to galaxies, from molecules to markets, from information to energy.

That previous generations did not see this isomorphism has two causes. First, the terminological walls between disciplines—physics calls it "action," chemistry calls it "free energy," economics calls it "equilibrium price," information science calls it "channel capacity," energy science calls it "efficiency limit." They use different names, but these names point to the same mathematical structure. Second, and more profoundly—calling $\delta S = 0$ "unified pricing" sounds too much like Aristotle's "final cause." Newton, after writing "Nature seems to always move along the path of least action" in the margin, crossed it out himself. Fermat, because Descartes said his principle of least time "sounded too much like light making choices," remained silent for twenty-five years. Einstein drew countless question marks next to Λ, finally calling it his "greatest blunder"—and then the constant he crossed out was proven to be the simplest model of dark energy. What previous generations dared not say aloud, this paper says for them.

8 Precise Positioning in the History of Ideas

In the history of human thought, there have been multiple grand unification attempts. Each touched upon some aspect of the universe's underlying algorithm from a certain dimension, but each stopped somewhere.

Pythagoras's "All things are numbers"—he gave the constraint conditions: numerical ratios are the simplest encoding of constraint conditions. But he did not know what the extremum criterion was. "All things are numbers" is only half the equation—having constraint conditions without an extremum criterion.

Aristotle's categories and logic—he gave the grammar: categories and logic stipulate how to speak a correct sentence. But he did not see that the grammar itself points to a single subject. Li in his categories is practical wisdom, in his logic the doctrine of the mean, in his poetics catharsis—but he did not recognize them as the same thing.

Leibniz's universal characteristic—he saw the need for a formal language to describe the universe's underlying algorithm. But he thought that language was a symbol

system invented by humans. It is not. It is built into the universe itself; humans have merely translated it from the independent languages of dozens of disciplines into a comprehensible unified grammar.

Spencer's Social Darwinism—he used "survival of the fittest" to connect biology and society. But his theory was empty analogy, lacking a precise mathematical criterion such as the evolutionarily stable strategy (ESS). Without an extremum criterion, "survival of the fittest" is merely a slogan, an empty shell that can be filled with any ideology. Li's framework provides the precision that Spencer lacked: ESS is Li's criterion in biology, market clearing is Li's criterion in economics. What Spencer attempted but failed to connect, Li's framework accomplishes with rigorous mathematical correspondence.

Bertalanffy's General System Theory—he proposed that systems share cross-disciplinary common structures. But he had only a framework, without axioms, without a driving force. He did not know that systems at every instant pursue the unique price at which a transaction can be reached under constraint conditions. His framework is a clock, but without a mainspring. Li is that mainspring.

The Vienna Circle's Logical Positivism—they proposed the core standard that science requires falsifiability. But their verifiability principle was itself unverifiable: the verifiability principle as a meta-statement—"all meaningful statements must be empirically verifiable"—is itself neither a logical truth nor an empirically verifiable statement, and is therefore meaningless by its own standard. This is an instantiation of Gödel's incompleteness theorem in epistemology. Then Gödel's incompleteness theorem collided with them—any sufficiently powerful self-consistent system has truths that cannot be proven internally. Their dream shattered on self-reference, but they never honestly admitted it. They simply changed the topic.

The fundamental difference between Li's unification framework and all these attempts is this: it is not a unification accomplished within any single discipline, not a connection of different disciplines through loose analogies, not merely philosophical speculation without a mathematical skeleton. It independently derives the same pricing grammar within each rigorously tested discipline, using that discipline's own mathematical framework—constraints, bargaining, the price at which a transaction is finally reached—and then proves that these derivations are identical in mathematical structure. Each disciplinary mapping contains that discipline's own definitions, theorems, and formulas, and can be independently reviewed and falsified by professional readers of that discipline.

Furthermore, Li's framework is the first system to honestly mark its boundary after completing cross-disciplinary unification. At the limit of knowledge, it does not declare "I have conquered all," but marks "Here is the boundary we cannot cross." In the history of human thought, all previous grand unification attempts—from Pythagoras to the Vienna Circle—either failed to achieve formalization, or shattered before Gödel's incompleteness theorem without ever honestly marking their own boundaries. Li is the first system to place a boundary stone at the frontier.

9 The Boundary of the Theory and the Honest Marking of Ontological Conviction

Li's unification framework can explain why all things obey the unified pricing rule—because this is the universe's most fundamental grammar. But it cannot answer a more fundamental question: Where does the unified pricing rule itself come from? Where does

"Li" come from?

To answer this question would require a meta-pricing rule more fundamental than "Li." And if such a meta-pricing rule existed, it would inevitably face the same question—where does its rule come from? This leads to infinite regress or a self-referential paradox. The logical structure of this boundary is rigorously given by Gödel's incompleteness theorem.

Gödel [20] proved: any consistent formal system T containing arithmetic and recursively axiomatizable necessarily contains a statement G such that $T \nvdash G$ and $T \nvdash \neg G$. Turing [21] proved: no universal algorithm exists that can determine whether an arbitrary Turing machine will halt on arbitrary input. Von Neumann [16] proved: any sufficiently complex self-reproducing system, whose internal error rate grows exponentially with the number of components, necessarily undergoes total collapse at some unpredictable critical point.

These three theorems together set an insurmountable boundary for the unified pricing algorithm. Any sufficiently powerful self-consistent system cannot completely price all of its own truths internally. The unified pricing rule can price all things, but cannot price itself. Where does "Li" come from—this question cannot be answered within any system. It is not that we have not yet found the answer; it is that the answer is logically impossible to find within the system. Gödel's incompleteness theorem, Turing's halting problem, and von Neumann's complexity threshold independently derived the same boundary in logic, computation theory, and engineering respectively. The fact that the self-referential structure was independently priced three times in three different disciplines is itself the final instantiation of Li's unified pricing grammar at the meta-level.

In the process of constructing this unification framework, the author of this paper underwent a cognitive transformation. Initially, the author thought he had merely discovered that all things "conform to" the same rule—Li in physics is the minimization of action, Li in chemistry is the minimization of free energy, Li in biology is the maximization of gene replication efficiency, Li in economics is the maximization of utility. But as verification deepened—from quantum mechanics to chemistry, from biology to economics, from information science to cosmology—the precision and universality of this "conformity" pointed to a deeper possibility: it is not that all things happen to conform to the description of Li, but rather that precisely because all things are themselves the unfolding of Li, all things necessarily conform to the description of Li.

Conformity is not coincidence. Conformity is not a constructed mapping. Conformity is a discovered ontological fact. From "all things conform to the description of Li" to "all things are themselves the unfolding of Li"—this is the reversal of the direction of causality from epistemology to ontology.

However, this paper must honestly mark: this ontological assertion can never be definitively proven. Between "conform to" and "are themselves" lies a logical leap—equating the optimal descriptive framework currently found by humans with the built-in code of the universe itself. This leap cannot be proven within the system, because the proof itself would require stepping outside the system. Gödel's incompleteness theorem sets an insurmountable logical boundary for this leap.

In the arena of science, this paper's theory remains at "all things conform to the description of Li"—this is a scientific proposition that can be tested, reviewed, and falsified. If any disciplinary mapping is found by mainstream researchers of that discipline to contain fundamental misunderstandings or forced correspondences, the entire system will collapse. This is the risk that a genuine grand unification must bear—and it is also its most honest quality.

Under the stars of philosophy, this paper retains the ontological conviction that "all things are themselves the unfolding of Li"—this is the truth that the theory whispers in solitude, the deepest insight reached by the theory builder after long, solitary inquiry. But conviction is not a proven theorem. This paper does not need to choose between the two—it need only honestly mark: here, science reaches its boundary; there, conviction takes up the torch.

The highest is not conquest; the highest is honesty.

10 Conclusion

This paper has demonstrated that gravity and electromagnetism do not need to be unified as the same kind of force—they are already two independent instantiations of the same pricing algorithm. Constraint conditions determine the space of possible states; the extremum criterion selects the unique realization from among possible states. Gravity is the process by which spacetime geometry prices its own curvature under constraint conditions; electromagnetism is the process by which the electromagnetic field prices its own field configuration under constraint conditions. What they share is not the essence of force, but the pricing grammar of "gauge connection $\rightarrow$ curvature $\rightarrow$ extremum criterion $\rightarrow$ field equation."

This grammar is not a coincidence internal to physics. This paper has further demonstrated that classical mechanics, quantum mechanics, cosmology, chemical equilibrium and non-equilibrium thermodynamics, market clearing and financial collapse, channel capacity and complexity threshold, and the entropy increase principle—the core pricing rules of these disciplines were independently derived throughout history, by researchers unaware of each other's work, using different mathematical tools, yet ultimately converged on the same constrained extremization skeleton in mathematical structure. Independently derived isomorphism is more convincing than intentionally inherited isomorphism.

Previous generations did not see this isomorphism, not because mathematical tools were insufficient, but because of the terminological walls between disciplines, and the fear of calling $\delta S = 0$ "unified pricing"—it sounded too much like final cause. Newton wrote it in the margin and crossed it out; Fermat remained silent for twenty-five years; Einstein drew countless question marks next to Λ. This paper takes up the truth they dared not speak: all things have no choice; constraints determine the unique solution.

In the history of human thought, there have been multiple grand unification attempts—from Pythagoras to the Vienna Circle. Each touched upon some aspect of the universe's underlying algorithm from a certain dimension, but each stopped somewhere. Some stopped at philosophical speculation, lacking a mathematical skeleton; some were accomplished within physics alone, unable to cross disciplinary boundaries; some relied on loose analogies, lacking rigorous formal mapping; some shattered before Gödel's incompleteness theorem, yet never honestly marked their own boundaries. Li's unification framework is the first system to place a boundary stone at the frontier after completing cross-disciplinary unification.

And at the limit of knowledge, the unified pricing rule can price all things, but cannot price itself. This is the logical boundary of the universe's unified pricing algorithm—not a failure, but honesty. The highest is not conquest; the highest is honesty.

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A Table of Symbols

Table 2: Symbols and their meanings.

Symbol	Meaning
S	Action functional
$\mathcal{L}$	Lagrangian density
ϕ	Field variable
$\mathcal{M}$	Base manifold
$\delta S = 0$	Extremum criterion
$g_{\mu\nu}$	Metric tensor
R	Ricci scalar
κ	Einstein gravitational constant $8\pi G/c^4$
$T_{\mu\nu}$	Energy-momentum tensor
A_μ	Electromagnetic four-potential
$F_{\mu\nu}$	Electromagnetic field tensor
J^μ	Four-current density
H	Hubble parameter
ρ	Matter density
Λ	Cosmological constant
G	Gibbs free energy
σ	Entropy production rate
C	Channel capacity
B	Bandwidth
S/N	Signal-to-noise ratio
N_{crit}	Complexity threshold
η	Heat engine efficiency
$\hbar$	Reduced Planck constant

B Fundamentals of the Calculus of Variations

For a functional $J[y] = \int_{x_1}^{x_2} L(x, y, y')\,dx$, its extremal curve satisfies the Euler-Lagrange equation:

$$\frac{\partial L}{\partial y} - \frac{d}{dx}\left(\frac{\partial L}{\partial y'}\right) = 0. \tag{8}$$

This is the common mathematical source of all physical field equations in this paper. In field theory, the Lagrangian density $\mathcal{L}$ replaces L, the field variable ϕ replaces y, and

functional derivatives replace ordinary derivatives, but the structure of the extremum criterion $\delta S = 0$ remains unchanged.

The Euler-Lagrange equation is a necessary condition for a functional to take an extremum. Regarding sufficient conditions for extrema, Hilbert established a systematic theory: when the Lagrangian L satisfies certain convexity conditions, the Hilbert invariant integral along the extremal curve is path-independent, and the extremal curve makes the functional take a strict minimum. In physics, the principle of least action in classical mechanics usually takes a minimum (brachistochrone, geodesics, etc.), but not all extrema in physical systems are minima—under certain constraint conditions, extrema may be saddle points. The core argument of this paper is based on the extremum criterion $\delta S = 0$, and does not depend on whether the extremum is a minimum, maximum, or saddle point. The constraint conditions select from all possible configurations the configuration that makes the first variation of the functional vanish—this configuration is the unique output of the pricing algorithm.